

Peiyang Song

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📁 [peiyang-song.github.io](https://github.com/peiyang-song)

Education

8/2026 – now **Carnegie Mellon University** Pittsburgh, PA
Ph.D. in Artificial Intelligence @ Language Technologies Institute (LTI)

9/2022 – 6/2026 **California Institute of Technology** Pasadena, CA
B.S. in Computer Science & Minor in Robotics
Advisors: Prof. Steven Low & Prof. Günter Niemeyer. GPA: **4.2/4.0**.

Research Interests

Formal Reasoning · Natural Language Reasoning · Neuro-Symbolic AI · AI4MATH · LLMs

Selected Publications & Preprints

Full list on Google Scholar: <https://scholar.google.com/citations?user=E1j11NQAAAAJ>

- Under Review **Interactive Evaluation Requires a Design Science**
Keyang Xuan*, Peiyang Song*, Pan Lu, Pengrui Han, Wenkai Li, Zhenyu Zhang, Zexue He, Wenyue Hua, Manling Li, Jiaxuan You, Adrian Weller, Yizhong Wang[†], and Jiaxin Pei[†] (* Equal Contribution, [†] Equal Advising)
ICML Workshop on Combining Theory and Benchmarks (CTB), 2026
- Under Review **Rethinking Psychometric Evaluation of LLMs: When and Why Self-Reports Predict Behavior**
Rafal D. Kocielnik, Pengrui Han, Peiyang Song, Myrl G. Marmarelis, Ramit Debnath, Dean Mobbs, Anima Anandkumar, R. Michael Alvarez
ICML Workshop on Combining Theory and Benchmarks (CTB), 2026, Best Paper Award
- Under Review **Characterizing Initial Human-AI Proof Formalization Workflows**
Katherine M. Collins*, Simon Frieder*, Jonas Bayer, Jacob Loader, Jeck Lim, Peiyang Song, Fabian Zasier, Lexin Zhou, Shanda Li, Sam Looi, Jose Hernandez-Orallo, Joshua B. Tenenbaum, Cameron Freer, Umang Bhatt, Adrian Weller, Valerie Chen[†], Ilia Sucholutsky[†] (* Equal Contribution, [†] Equal Advising)
NeurIPS Workshop on Mathematical Reasoning and AI (MATH-AI), 2025
- Under Review **Steer2Adapt: Dynamically Composing Steering Vectors Elicits Efficient Adaptation of LLMs**
Pengrui Han*, Xueqiang Xu*, Keyang Xuan*, Peiyang Song, Siru Ouyang, Runchu Tian, Yuqing Jiang, Qian Cheng, Pengcheng Jiang, Jiashuo Sun, Junxia Cui, Ming Zhong, Ge Liu, Jiawei Han, Jiaxuan You (* Equal Contribution)

- TMLR 2026 **Large Language Model Reasoning Failures**
Peiyang Song*, Pengrui Han*, Noah Goodman (* Equal Contribution)
Transactions on Machine Learning Research, 2026, Survey Certification
- ICML 2026 **The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs**
Pengrui Han*, Rafal D. Kocielnik*, Peiyang Song, Ramit Debnath, Dean Mobbs, Anima Anandkumar, R. Michael Alvarez (* Equal Contribution)
International Conference on Machine Learning, 2026
Best Paper Honorable Mention @ NeurIPS LAW Workshop, 2025
- NeuS 2025 **Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean**
Peiyang Song, Kaiyu Yang, Anima Anandkumar
International Conference on Neuro-symbolic Systems, 2025
1.3k+ stars on Github, ranking 2nd after Mathlib4 among all Lean projects
- IEEE Micro 2025 **Delay Space Arithmetic and Architecture**
Rhys Gretsch, Peiyang Song, Advait Madhavan, Jeremy Lau, Timothy Sherwood
IEEE Micro, 2025, Top Pick Award
- ICLR 2025 **LeanAgent: Lifelong Learning for Formal Theorem Proving**
Adarsh Kumarappan*, Mo Tiwari*, Peiyang Song, Robert Joseph George, Chaowei Xiao, Anima Anandkumar
International Conference on Learning Representations, 2025
- TMLR 2025 **LeanProgress: Guiding Search for Neural Theorem Proving via Proof Progress Prediction**
Robert Joseph George, Suozhi Huang, Peiyang Song, Anima Anandkumar
Transactions on Machine Learning Research, 2025
- In Submission **Adaptation of Agentic AI: A Survey of Post-Training, Memory, and Skills**
Pengcheng Jiang*, Jiacheng Lin*, Zhiyi Shi*, Zifeng Wang, Luxi He, Yichen Wu, Ming Zhong, Peiyang Song, Qizheng Zhang, Heng Wang, Xueqiang Xu, Hanwen Xu, Pengrui Han, Dylan Zhang, Jiashuo Sun, Chaoqi Yang, Kun Qian, Tian Wang, Changran Hu, Manling Li, Quanzheng Li, Hao Peng, Sheng Wang, Jingbo Shang, Chao Zhang, Jiaxuan You, Liyuan Liu, Pan Lu, Yu Zhang, Heng Ji, Yejin Choi, Dawn Song, Jimeng Sun, Jiawei Han (* Equal Contribution)
arXiv Preprint, 2025
- Under Review **How and Why LLMs Generalize: A Fine-Grained Analysis of LLM Reasoning from Cognitive Behaviors to Low-Level Patterns**
Haoyue Bai, Yiyue Sun, Wenjie Hu, Shi Qiu, Maggie Ziyu Huan, Peiyang Song, Robert Nowak, Dawn Song
arXiv Preprint, 2025

- EMNLP 2024 **In-Context Learning May Not Elicit Trustworthy Reasoning: A-Not-B Errors in Pretrained Language Models**
Pengrui Han*, Peiyang Song*, Haofei Yu, Jiaxuan You (* Equal Contribution)
Conference on Empirical Methods in Natural Language Processing: Findings, 2024
- EMNLP 2024 **Creative and Context-Aware Translation of East Asian Idioms with GPT-4**
Kenan Tang*, Peiyang Song*, Yao Qin, Xifeng Yan (* Equal Contribution)
Conference on Empirical Methods in Natural Language Processing: Findings, 2024
- ASPLOS 2024 **Energy Efficient Convolution with Temporal Arithmetic**
Rhys Gretsch, Peiyang Song, Advait Madhavan, Jeremy Lau, Timothy Sherwood
ACM International Conference on Architectural Support for Programming Languages and Operating Systems, 2024
- NeurIPS 2023 **LeanDojo: Theorem Proving with Retrieval-Augmented Language Models**
Kaiyu Yang, Aidan Swope, Alex Gu, Rahul Chalamala, Peiyang Song, Shixing Yu, Saad Godil, Ryan Prenger, Anima Anandkumar
Conference on Neural Information Processing Systems, 2023, Oral Presentation

Selected Academic Awards

- 7/2026 **Best Paper Award @ ICML CTB Workshop**
For paper “Rethinking Psychometric Evaluation of LLMs: When and Why Self-Reports Predict Behavior”
- 5/2026 **Outstanding Reviewer Award @ EACL**
In recognition of outstanding review service at EACL 2026
- 1/2026 **Survey Certification @ TMLR**
For paper “Large Language Model Reasoning Failures”
- 12/2025 **Best Paper Honorable Mention Award @ NeurIPS LAW Workshop**
For paper “The Personality Illusion: Revealing Dissociation Between Self-Reports & Behavior in LLMs”
- 5/2025 **ICLR Notable Reviewer Award**
In recognition of outstanding review service at ICLR 2025
- 4/2025 **George W. Housner Student Discovery Fund**
For paper “Lean Copilot: Large Language Models as Copilots for Theorem Proving in Lean”
- 2/2025 **IEEE Micro Top Pick Award**
For paper “Energy Efficient Convolutions with Temporal Arithmetic”
- 4/2023 **Caltech SURF Fellowship**
For research on machine learning for theorem proving in Lean

Selected Media

- 2026 **'Not how you build a digital mind': How reasoning failures are preventing AI models from achieving human-level intelligence**
Live Science
- 2026 **Scientists Found AI's Fatal Flaw—The Most Advanced Models Are Failing Basic Logic Tests**
Popular Mechanics
- 2025 **New Framework Simplifies the Complex Landscape of Agentic AI**
VentureBeat
- 2025 **This AI Paper Explains Why Most “Agentic AI” Systems Feel Impressive in Demos and then Completely Fall Apart in Real Use**
MarkTechPost
- 2025 **Researchers Discover “Personality Illusion” to Reveal a Profound Disconnect Between Language and Behavior in LLMs**
MIT Technology Review China
- 2024 **Mathematicians' Newest Assistants Are Artificially Intelligent**
Scientific American
- 2024 **LeanAgent: The First Life-Long Learning Agent for Formal Theorem Proving in Lean**
MarkTechPost
- 2024 **Lean Copilot: An AI Tool That Allows Large Language Models (LLMs) to Be Used in Lean for Proof Automation**
MarkTechPost
- 2023 **Can LLMs Generate Mathematical Proofs That Can Be Rigorously Checked?**
MarkTechPost

Invited Talks

Large Language Model Reasoning Failures

- 3/2026 Fireside Chat @ Ploutos.ai
- 3/2026 BlueDot AI Safety Reading Group

Towards An AI Mathematician

- 12/2023 UC Santa Barbara NLP Lab
- 11/2023 CCS Research & Creative Activities Conference (RACA-CON)
- 8/2023 Caltech SURF Seminar Day

Teaching Experience

- Spring 2026 **CS 38/138: Algorithms**
Teaching Assistant @ *California Institute of Technology*
- Winter 2026 **ME/CS/EE 133B: Robotics – Planning**

Teaching Assistant @ *California Institute of Technology*
Fall 2025 **ME/CS/EE 133A: Robotics – Kinematics**
Teaching Assistant @ *California Institute of Technology*

Academic Services

Organizer The 6th Workshop on Mathematical Reasoning and AI (MATH-AI) @ NeurIPS 2026

Tutorial Instructor Tutorial on Neuro-Symbolic Theorem Proving with Lean @ 3rd Neuro-Symbolic AI Summer School (NSSS)

Reviewer Conference on Neural Information Processing Systems (NeurIPS)
International Conference on Learning Representations (ICLR)
Conference on Language Modeling (COLM)
ACM Computing Surveys
Association for Computational Linguistics Rolling Review (ARR)
Annual Meeting of the Association for Computational Linguistics (ACL)
Conference on Empirical Methods in Natural Language Processing (EMNLP)
Nations of the Americas Chapter of the Association for Computational Linguistics (NAACL)
European Chapter of the Association for Computational Linguistics (EACL)
Asia-Pacific Chapter of the Association for Computational Linguistics (AACL)
International Joint Conference on Natural Language Processing (IJCNLP)

NeurIPS Mathematical Reasoning and AI (MATH-AI) Workshop
NeurIPS Workshop on Deep Learning for Code (DL4C)
NeurIPS Workshop on Behavioral Machine Learning
ICLR VeriAI: AI Verification in the Wild Workshop
ICLR Workshop on Representational Alignment (Re-Align)
ICML AI for Math (AI4MATH) Workshop
ICML Workshop on LLMs and Cognition (LLM-Cognition)
ICML Workshop on Assessing World Models
ICML Workshop on Models of Human Feedback for AI Alignment (MoFA)
COLM Workshop on Agent Behavior (WAB)

On-Campus Services & Appointments

Caltech Admissions Ambassador @ Caltech Undergraduate Admissions Office
First-Year Caltech Connector (FCC) @ Student & Family Engagement Office